



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Programs

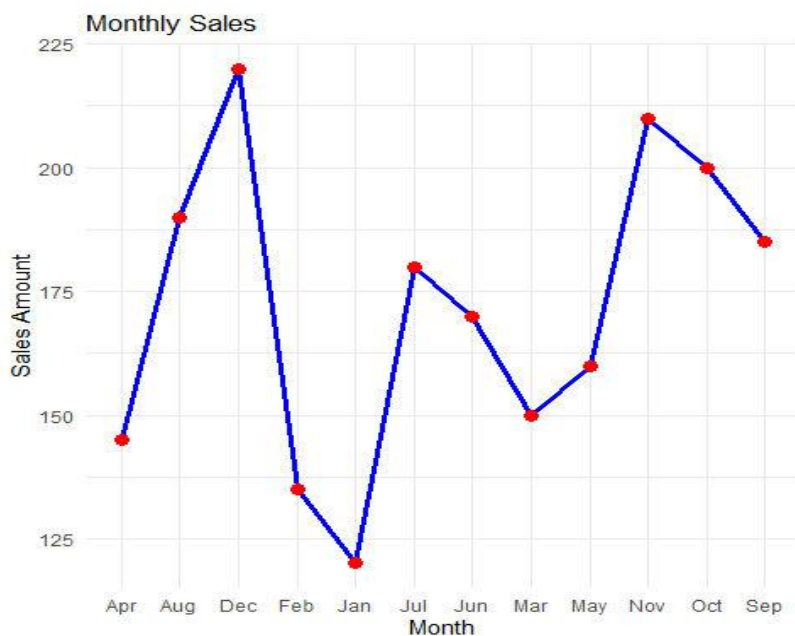
SET 1-Monthly Sales Data

1A. Line Chart

Program:

```
ggplot(sales_data,aes(x=Month,y=Sales,group=1))+  
geom_line(color="blue",linewidth=1.2)+  
geom_point(color="red",size=3)+  
labs(title="Monthly Sales",x="Month",y="Sales Amount")+  
theme_minimal()
```

Output:



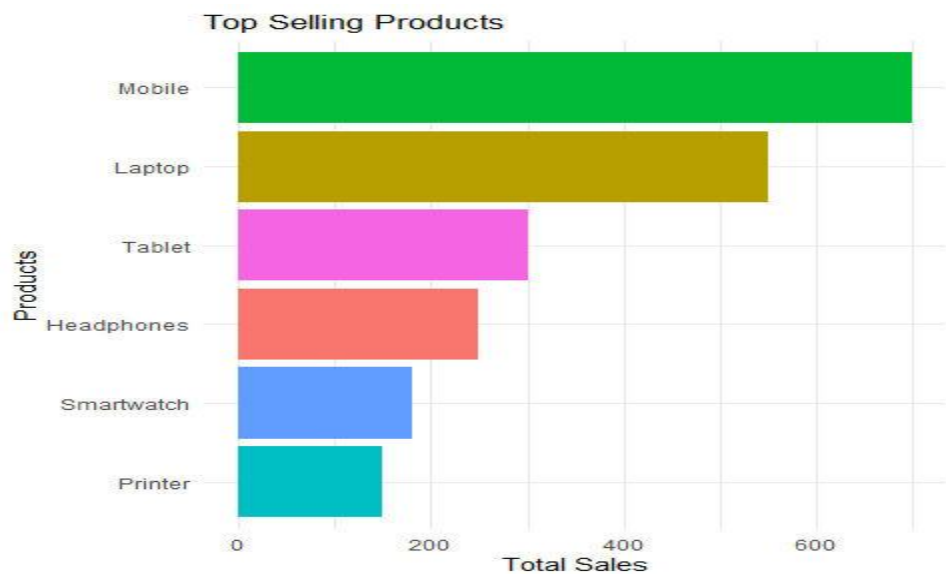
1B. Bar Chart

Program:

```
ggplot(product_data,aes(x=reorder(Product,Sales),y=Sales,fill=Product))+  
geom_bar(stat="identity")+  
coord_flip()+  
labs(title="Top Selling Products",x="Products",y="Total Sales")+
```

```
theme_minimal()+
theme(legend.position="none")
```

Output:



1C. Scatter Plot

Program:

```
ggplot(sales_data,aes(x=Advertising_Budget,y=Sales))+
geom_point(size=4,color="darkgreen")+
geom_smooth(method="lm",se=FALSE,color="red")+
labs(title="Advertising Budget vs Monthly Sales",
x="Advertising Budget",
y="Monthly Sales")+
theme_minimal()
```

Output:

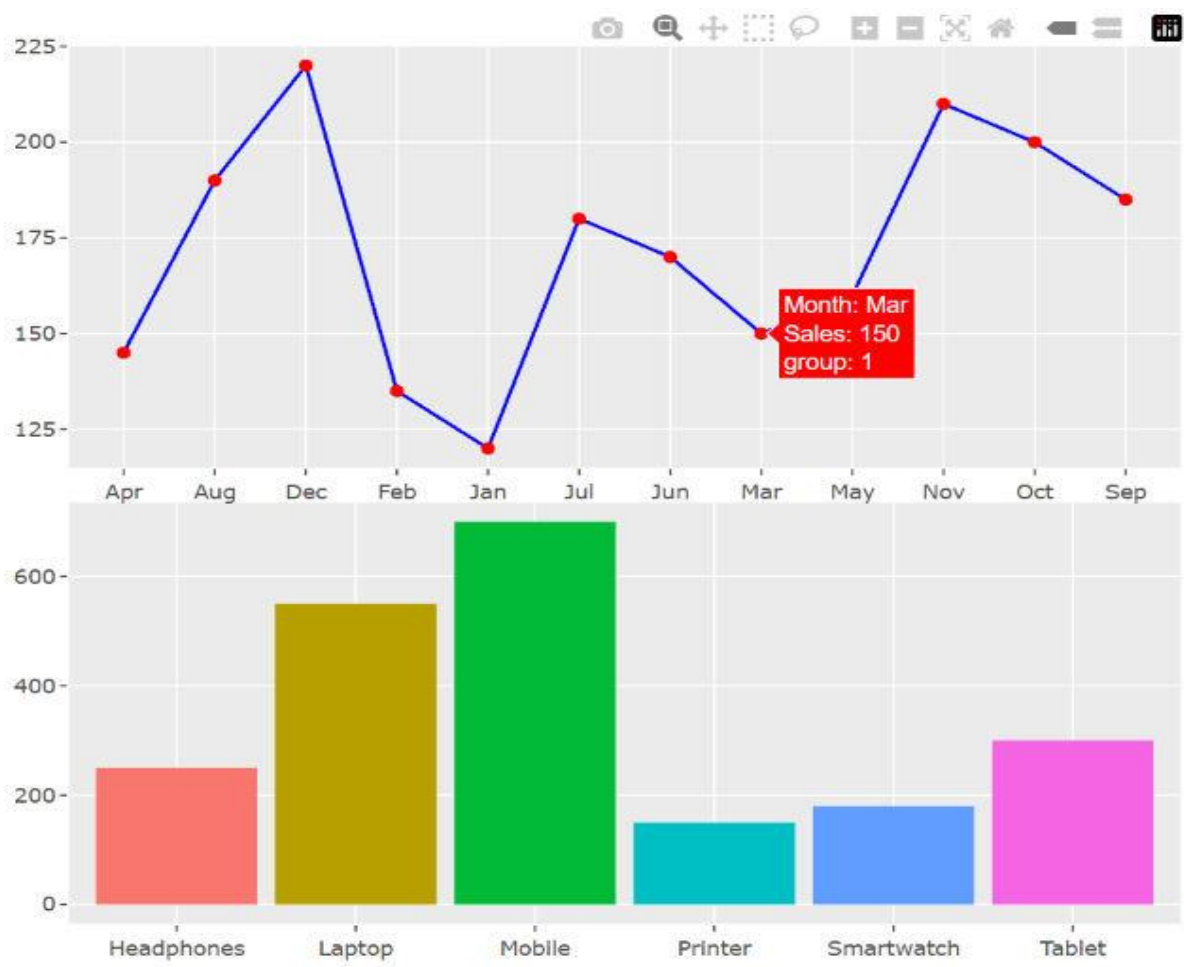


1D. Combined Interactive Dashboard

Program:

```
line_plot <- ggplotly(  
  ggplot(sales_data,aes(x=Month,y=Sales,group=1))+  
  geom_line(color="blue")+  
  geom_point(color="red")  
)  
bar_plot <- ggplotly(  
  ggplot(product_data,aes(x=Product,y=Sales,fill=Product))+  
  geom_bar(stat="identity")+  
  theme(legend.position="none")  
)  
subplot(line_plot,bar_plot,nrows=2)
```

Output:



SET 2- Customer Feedback Analysis

2A. Histogram of Customer Ages

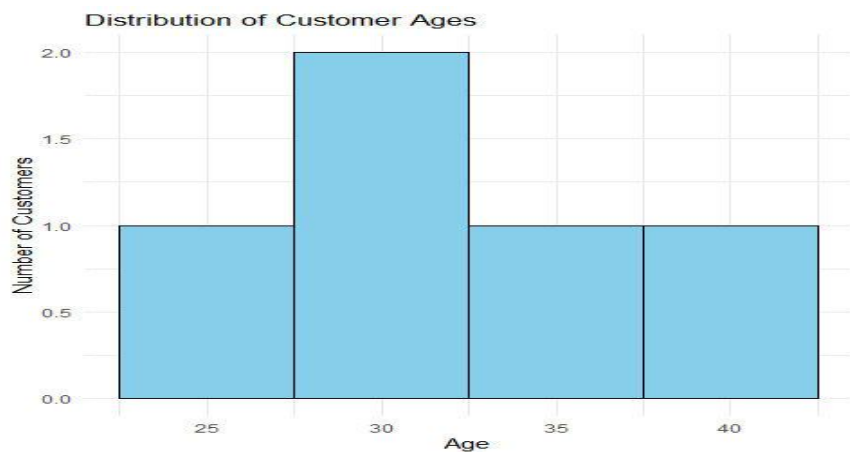
Step 1: Install Packages (Run Only Once)

```
install.packages("ggplot2")  
install.packages("dplyr")  
install.packages("wordcloud")  
install.packages("RColorBrewer")
```

Program:

```
library(ggplot2)  
customer_data <- data.frame(  
  CustomerID=c(1,2,3,4,5),  
  Age=c(25,30,35,28,40),  
  SatisfactionScore=c(4,5,3,4,5)  
)  
  
ggplot(customer_data, aes(x=Age)) +  
  geom_histogram(binwidth=5, fill="skyblue", color="black") + labs(  
    title="Distribution of Customer Ages",  
    x="Age",  
    y="Number of Customers"  
  ) +  
  theme_minimal()
```

Output:



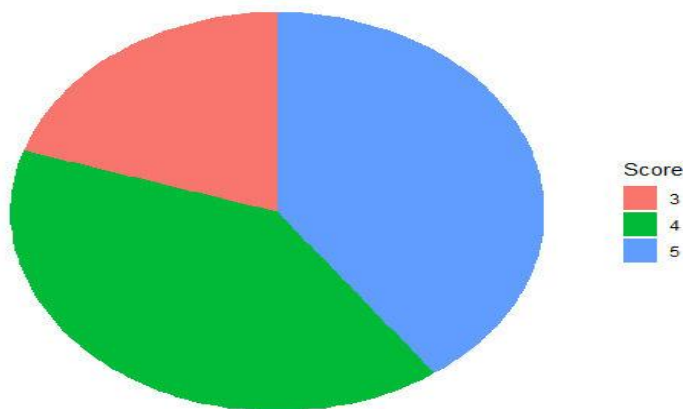
2B. Pie Chart of Satisfaction Scores

Program:

```
score_count <- customer_data %>%  
count(SatisfactionScore)  
  
ggplot(score_count,aes(x="",y=n,fill=factor(SatisfactionScore)))+  
geom_bar(stat="identity",width=1)+  
coord_polar("y")+  
labs(title="Customer Satisfaction Scores",  
fill="Score")+  
theme_void()
```

Output:

Customer Satisfaction Scores



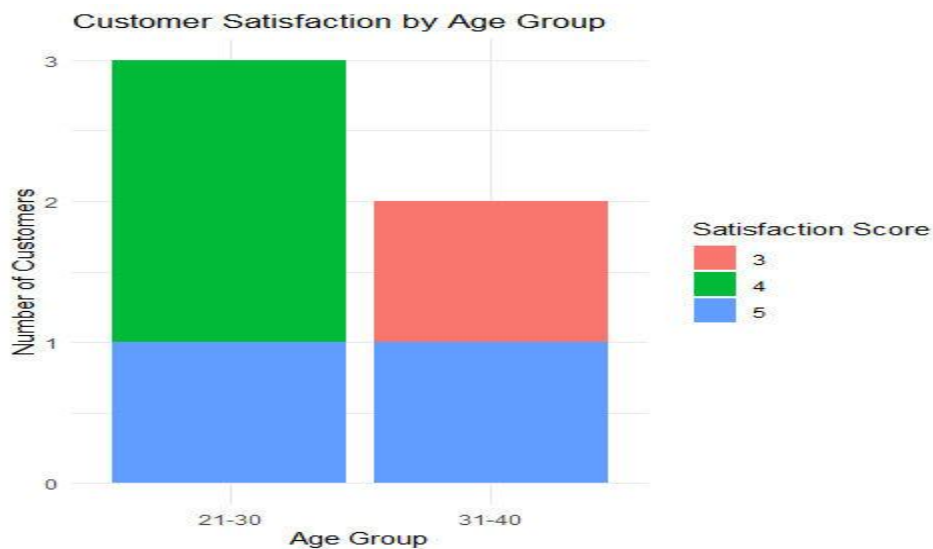
2C. Stacked Bar Chart by Age Group

Program:

```
customer_data$AgeGroup <- cut(customer_data$Age,  
breaks=c(20,30,40,50),  
labels=c("21-30","31-40","41-50"),  
include.lowest=TRUE)  
  
customer_data  
  
ggplot(customer_data,  
aes(x=AgeGroup,  
fill=factor(SatisfactionScore)))+
```

```
geom_bar()+
labs(title="Customer Satisfaction by Age Group",
x="Age Group",
y="Number of Customers",
fill="Satisfaction Score")+
theme_minimal()
```

Output:



2D.Word Cloud

Program:

```
feedback <- c(
  "Excellent service and friendly staff",
  "Very satisfied with product quality",
  "Good customer support and fast delivery",
  "Excellent quality and quick response",
  "Very happy with the service"
)

docs <- Corpus(VectorSource(feedback))
docs <- tm_map(docs, content_transformer(tolower))
docs <- tm_map(docs, removePunctuation)
docs <- tm_map(docs, removeNumbers)
docs <- tm_map(docs, removeWords, stopwords("english"))
```

```
docs <- tm_map(docs, stripWhitespace)
dtm <- DocumentTermMatrix(docs)
m <- as.matrix(dtm)
word_freq <- sort(colSums(m), decreasing=TRUE)
wordcloud(
  names(word_freq),
  word_freq,
  min.freq=1,
  colors=brewer.pal(8,"Dark2"),
  random.order=FALSE,
  scale=c(4,1)
)
```

Output:



Set 3- Employee Performance Evaluation.

3A. Line Chart

Program:

```
employee_data <- data.frame(
  EmployeeID=c(1,2,3,4,5),
```

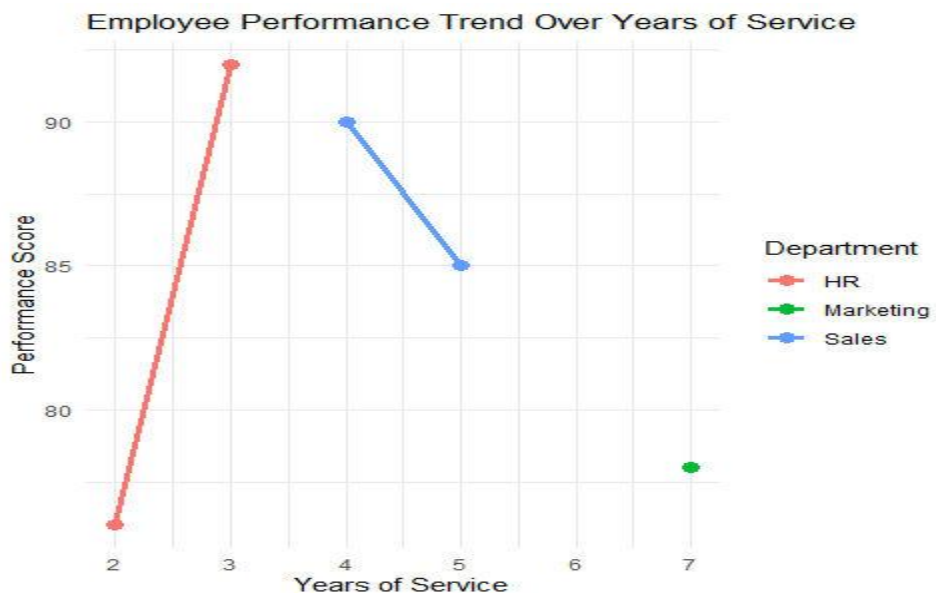
```

Department=c("Sales","HR","Marketing","Sales","HR"),
YearsOfService=c(5,3,7,4,2),
PerformanceScore=c(85,92,78,90,76)
)

ggplot(employee_data,
aes(x=YearsOfService,
y=PerformanceScore,
color=Department,
group=Department))+
geom_line(linewidth=1.2)+
geom_point(size=3)+
labs(title="Employee Performance Trend Over Years of Service",
x="Years of Service",
y="Performance Score",
color="Department")+
theme_minimal()

```

Output:



3B. Bar Chart (Department Distribution)

Program:

```

ggplot(employee_data,

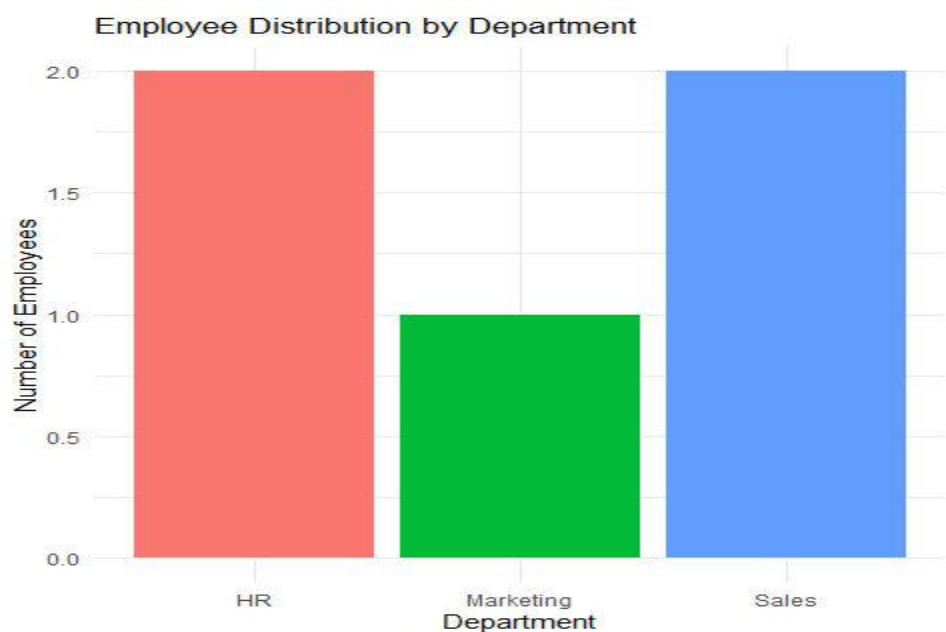
```

```

aes(x=Department,
fill=Department))+
geom_bar()+
labs(title="Employee Distribution by Department",
x="Department",
y="Number of Employees")+
theme_minimal()+
theme(legend.position="none")

```

Output:



3C. Scatter Plot (Years of Service vs Performance Score)

Program:

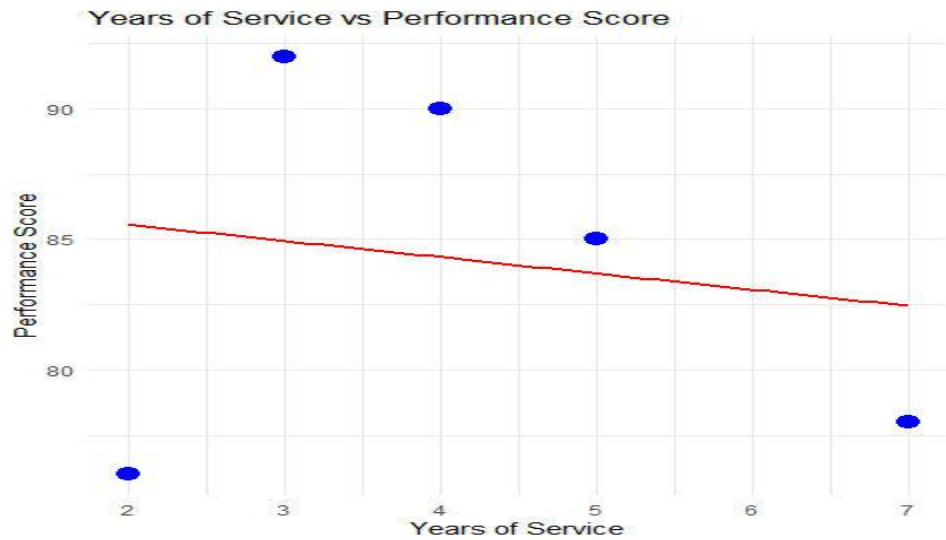
```

library(ggplot2)
ggplot(employee_data, aes(x=YearsOfService, y=PerformanceScore)) +
geom_point(size=4, color="blue") +
geom_smooth(method="lm", se=FALSE, color="red") +
labs(
title="Years of Service vs Performance Score",
x="Years of Service",
y="Performance Score"
) +

```

```
theme_minimal()
```

Output:



3D. Interactive Dashboard

Program:

```
library(plotly)
```

```
line_plot <- ggplotly(  
  ggplot(employee_data,  
    aes(x=YearsOfService,  
        y=PerformanceScore,  
        color=Department,  
        group=Department)) +  
  geom_line(size=1) +  
  geom_point(size=3) +  
  labs(title="Employee Performance Trend",  
        x="Years of Service",  
        y="Performance Score") +  
  theme_minimal()  
)
```

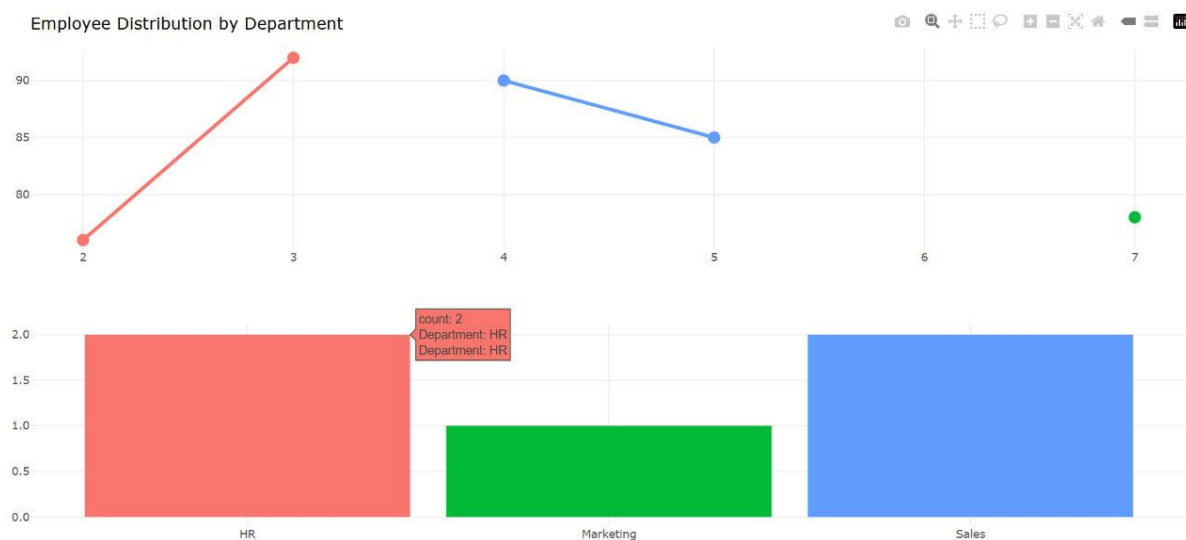
```
bar_plot <- ggplotly(  
  ggplot(employee_data,
```

```

aes(x=Department,
    fill=Department)) +
geom_bar() +
labs(title="Employee Distribution by Department",
     x="Department",
     y="Number of Employees") +
theme_minimal() +
theme(legend.position="none")
)
dashboard <- subplot(line_plot, bar_plot, nrow=2, margin=0.08)
dashboard

```

Output:



Set 4: Product Inventory Management

4A. Bar Chart

Program:

```

inventory_data <- data.frame(
  ProductID=c(1,2,3,4,5),
  ProductName=c("Product A","Product B","Product C","Product D","Product E"),
  QuantityAvailable=c(250,175,300,200,220)
)

```

```
ggplot(inventory_data,
aes(x=ProductName,
y=QuantityAvailable,
fill=ProductName))+
geom_bar(stat="identity")+
labs(title="Product Inventory",
x="Product Name",
y="Quantity Available")+
theme_minimal()+
theme(legend.position="none")
```

Output:



4B. Stacked Bar Chart

Program:

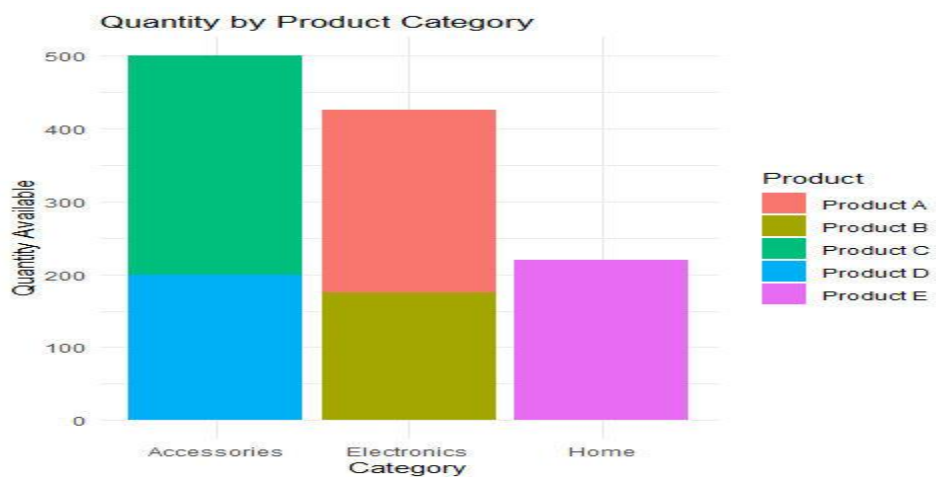
```
inventory_data$Category <- c(
"Electronics",
"Electronics",
"Accessories",
"Accessories",
"Home"
)
ggplot(inventory_data,
```

```

aes(x=Category,
y=QuantityAvailable,
fill=ProductName))+
geom_bar(stat="identity")+
labs(title="Quantity by Product Category",
x="Category",
y="Quantity Available",
fill="Product")+
theme_minimal()

```

Output:



4C. Scatter Plot

Program:

```

inventory_data$Price <- c(500,350,650,450,550)
ggplot(inventory_data,
aes(x=Price,
y=QuantityAvailable))+
geom_point(size=4,color="blue")+
geom_smooth(method="lm",se=FALSE,color="red")+
labs(title="Price vs Quantity Available",
x="Product Price",
y="Quantity Available")+
theme_minimal()

```

Output:



4D. Interactive Dashboard (R)

Program:

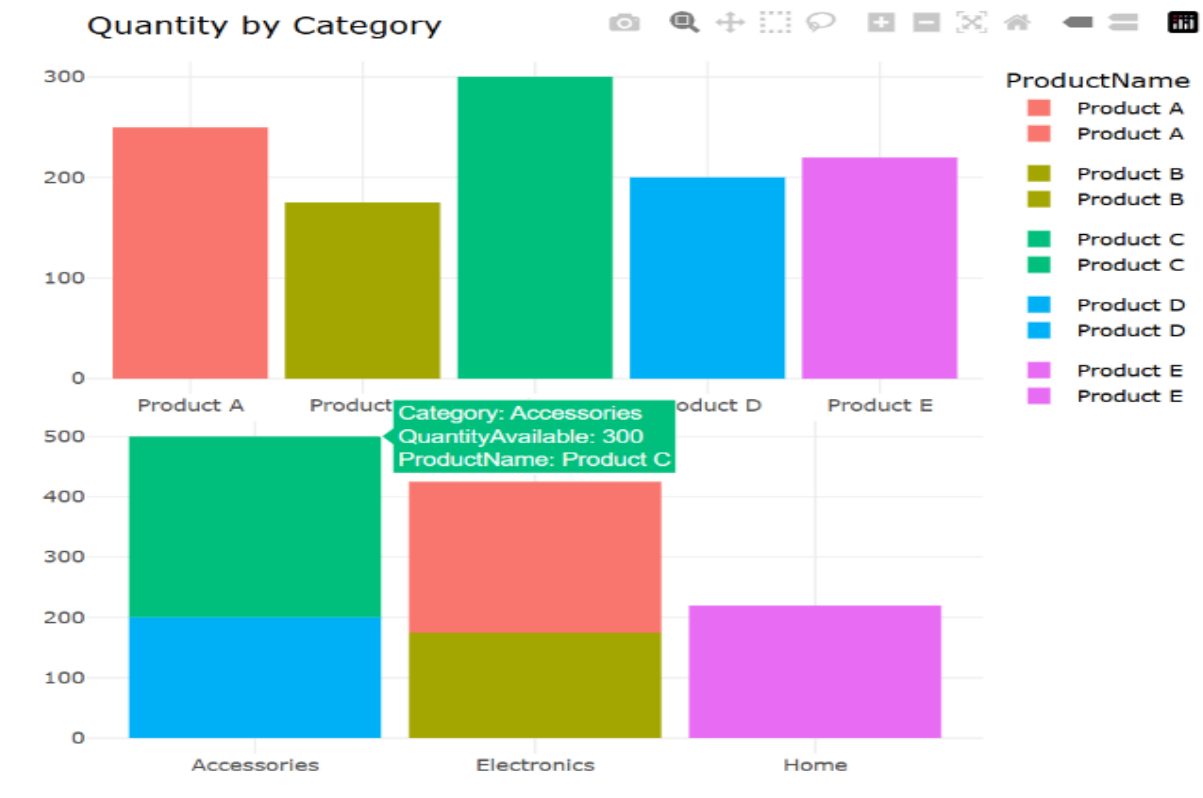
```
library(plotly)

bar_plot <- ggplotly(
  ggplot(inventory_data,
    aes(x=ProductName,
      y=QuantityAvailable,
      fill=ProductName))+
    geom_bar(stat="identity")+
    labs(title="Product Inventory")+
    theme_minimal()+
    theme(legend.position="none")
  )

stacked_plot <- ggplotly(
  ggplot(inventory_data,
    aes(x=Category,
      y=QuantityAvailable,
      fill=ProductName))+
    geom_bar(stat="identity")+
    labs(title="Quantity by Category")+
```

```
theme_minimal()
)
dashboard <- subplot(bar_plot, stacked_plot, nrow=2)
dashboard
```

Output:



SET 5- Website Analytics

5A. Line Chart

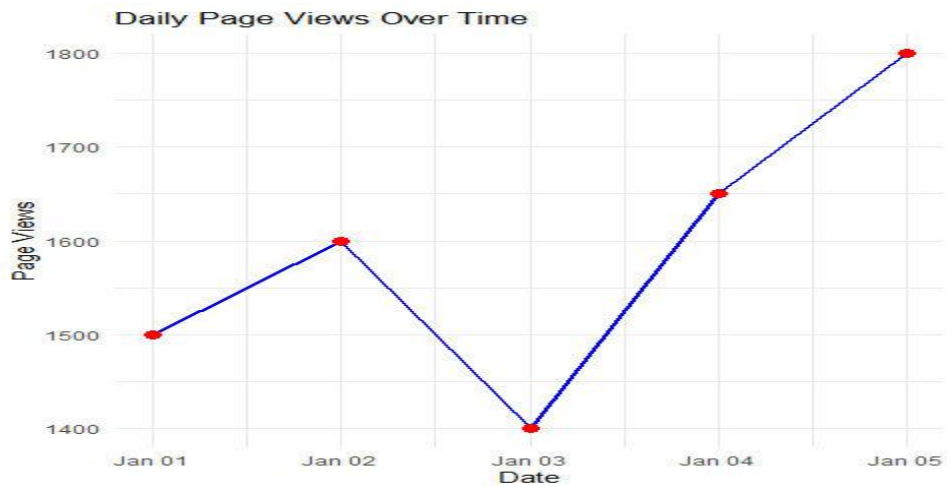
Program:

```
website_data <- data.frame(
  Date=as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  PageViews=c(1500,1600,1400,1650,1800),
  CTR=c(2.3,2.7,2.0,2.4,2.6)
)

ggplot(website_data,
  aes(x=Date,y=PageViews,group=1))+
  geom_line(color="blue",size=1)+
  geom_point(color="red",size=3)+
```

```
labs(title="Daily Page Views Over Time",
x="Date",
y="Page Views")+
theme_minimal()
```

Output:

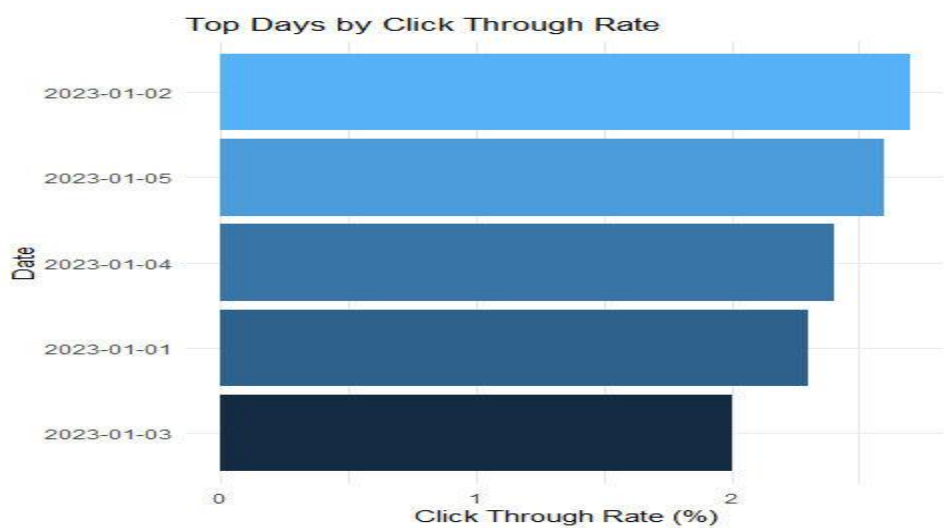


5B. Bar Chart (Top N Days with Highest CTR)

Program:

```
top_ctr <- website_data %>%
  arrange(desc(CTR))
ggplot(top_ctr,
  aes(x=reorder(as.character(Date),CTR),
  y=CTR,
  fill=CTR))+
  geom_bar(stat="identity")+
  coord_flip()+
  labs(title="Top Days by Click Through Rate",
  x="Date",
  ("%")+y="Click Through Rate (%)")+
  theme_minimal()+
  theme(legend.position="none")
```

Output:



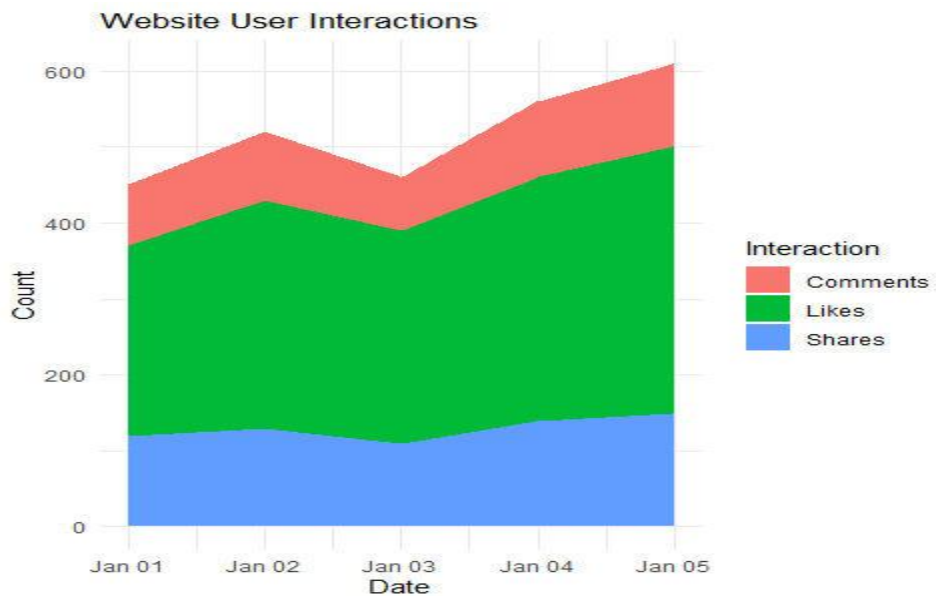
5C. Stacked Area Chart

Program:

```
website_data$Likes <- c(250,300,280,320,350)
website_data$Shares <- c(120,130,110,140,150)
website_data$Comments <- c(80,90,70,100,110)
interaction_data <- data.frame(
  Date=rep(website_data$Date,3),
  Interaction=c(rep("Likes",5),
    rep("Shares",5),
    rep("Comments",5)),
  Count=c(website_data$Likes,
    website_data$Shares,
    website_data$Comments)
)
ggplot(interaction_data,
  aes(x=Date,
    y=Count,
    fill=Interaction))+
  geom_area()+
  labs(title="Website User Interactions",
```

```
x="Date",
y="Count")+
theme_minimal()
```

Output:



5D. Interactive Dashboard

Program:

```
line_plot <- ggplotly(
  ggplot(website_data,
    aes(x=Date,y=PageViews,group=1))+
    geom_line(color="blue")+
    geom_point(color="red",size=3)+
    labs(title="Daily Page Views")
  )
bar_plot <- ggplotly(
  ggplot(top_ctr,
    aes(x=as.character(Date),y=CTR,fill=CTR))+
    geom_bar(stat="identity")+
    labs(title="Top CTR Days")+
    theme(legend.position="none")
  )
```

```
dashboard <- subplot(line_plot,bar_plot,nrows=2)
```

```
dashboard
```

Output:

